

Decent Coaching Classes

VIII SB

Mind Your Math - VII

Marks: 40

Time: 2 Hrs.

Instructions: All questions are compulsory (internal choice as indicated). Figures to the right indicate full marks. Draw neat figures wherever necessary.

Q.1 (A) Choose the correct alternative.

[5]

- (1) Which of the following is an irrational number?
(A) $\sqrt{16}$ (B) $\sqrt{25}$ (C) $\sqrt{7}$ (D) $\sqrt{49}$
- (2) Two parallel lines are intersected by a transversal. If one of the interior angles on the same side of the transversal measures 65° , the other measures _____.
(A) 65° (B) 115° (C) 25° (D) 125°
- (3) The cube root of 0.008 is _____.
(A) 0.02 (B) 2 (C) 0.002 (D) 0.2
- (4) The orthocentre of a right angled triangle is _____.
(A) the midpoint of the hypotenuse (B) the vertex of the right angle
(C) in the interior of the triangle (D) in the exterior of the triangle
- (5) The marked price of an article is ₹500. If a discount of 10% is given, the selling price is _____.
(A) ₹490 (B) ₹550 (C) ₹450 (D) ₹400

Q.1 (B) Solve the following.

[3]

- (1) Write $7/8$ in decimal form.
- (2) Write using the symbol of variation: "The area (A) of a circle is directly proportional to the square of its radius (r)."
- (3) Name the quadrilateral in which only one pair of opposite sides is parallel.

Q.2 Solve any four of the following.

[8]

- (a) Compare the rational numbers $-3/4$ and $-5/6$. Write using the proper symbol of $<$, $=$ or $>$.
- (b) Expand: $(m + 7)(m - 3)$
- (c) Factorise: $x^2 + 9x + 20$
- (d) Find the cube root of 3375.
- (e) The diagonal of a rectangle is 13 cm and one of its sides is 5 cm. Find the length of the other side.
- (f) An agent sold a plot worth ₹4,50,000 and charged 2% commission from the seller. Find the commission and the amount the seller received.

Q.3 Solve any four of the following.

[12]

- (a) Two parallel lines are intersected by a transversal. The measures of a pair of interior angles on the same side of the transversal are $(2x + 10)^\circ$ and $(3x + 20)^\circ$. Find the value of x and the measure of each angle.
- (b) Find the value of: $8^{2/3} + 16^{3/4}$
- (c) In $\triangle ABC$, seg AD and seg BE are medians and point G is the centroid. If $l(AG) = 10$ cm and $l(GE) = 4$ cm, find $l(GD)$ and $l(BE)$.
- (d) Expand: $(3x + 2y)^3$

- (e) Factorise: $2x^2 - 5x - 12$
- (f) x varies directly as y . When $x = 8$, $y = 20$. Find the constant of variation and the equation of variation. Hence find x when $y = 35$.

Q.4 Solve any three of the following.

[12]

- (a) Show the number $\sqrt{5}$ on a number line. Write the steps of the construction.
- (b) Construct $\square MORE$ such that $l(MO) = 5$ cm, $l(OR) = 4$ cm, $m\angle M = 75^\circ$, $m\angle O = 105^\circ$, $m\angle R = 90^\circ$.
- (c) Factorise: (i) $8x^3 + 125y^3$ (ii) $6x^2 - x - 15$
- (d) The marked price of a fan is ₹2,500. The shopkeeper gives 12% discount on it and still gets 10% profit. Find the cost price of the fan.
- (e) 12 workers can complete a piece of work in 15 days. Identify the type of variation and find how many workers are required to complete the same work in 9 days.